

Dotcast Uses Avnet Design Services' System Engineering Service to Speed Time-to-Market for Data Broadcast System

Avnet Design Services engineering team designs PC add-in card

Dotcast Inc. is a dynamic, privately held company based in Silicon Valley that is developing a national high-speed digital network for the distribution of digital entertainment, interactive services and multimedia communications.

The new network supports delivery of data and video services through the existing television distribution infrastructure, whether North American NTSC and DTV, or international PAL and DVB. Dotcast enables television broadcasters to accelerate delivery of new broadband digital services. As a result, cable companies can expand channel capacity to support interactive services and video on demand. Dotcast also provides rich-media content developers with an inexpensive, fast and reliable distribution path to the home.

To be successful, Dotcast had to get their PC-based datacasting board to market in the shortest possible time. Unfortunately, like most start-ups, they lacked the engineering design resources. "We were a one-year-old company without sufficient engineering support in a tight job market," said Vincent Le Chevalier of Engineering. "Our first boards had to be designed and built before we could demonstrate our capability to key partners in the broadcasting industry. We needed a design services partner who could help us through specification development, board level design, prototype manufacturing and test and quick ramp to production, all on a short time to market fuse."

"Avnet Design Services took a creative approach to the design of our board," said Vincent. "They had a reference design from one of their key

suppliers in the multimedia market, Philips Semiconductor, that they were able to customize for our applications by adding a Xilinx FPGA.

Their approach saved us several weeks and gave us a solid, proven hardware platform on which to overlay our content management software. Other design services organizations might have tried to start from scratch to increase their consulting revenue, but Avnet Design Services was motivated to get our design done quickly, since they want to sell us the components on the board. The quicker we get to market, the more successful we are and the more components Avnet sells us. It is a win-win proposition."

"Today we have an engineering team at Dotcast, but we still find it difficult to hire enough engineers to complete all the projects we need to keep our business growing," concluded Vincent. "We will continue to use Avnet Design Services to augment our engineering efforts and to assist us in building prototypes and transferring our volume production to manufacturing."

The Dotcast data-broadcasting PCI card plugs into a standard PC platform. It supports DTV and HDTV based data broadcasting (datacasting) to the PC. The board uses the Philips TDA89XX chipset to process video and audio signals. The system includes functions for DTV, audio decoding and mixing, a tuner and MPOC and SPDIF. A Xilinx FPGA was added to the design to provide a PCI Bridge function to control dNTSC encode and decode functions, JTAG interrupt control, and an expansion bus for increased flexibility.

"Avnet Design Services took a creative approach to the design of our board..."

*Vincent Le Chevalier,
Vice President of Engineering
Dotcast, Inc.*

North America
2211 South 47th St.
Phoenix, Arizona 85034
United States
Tel: +1 (800) 585-1602

Europe
79 Rue Pierre Sémeard BP90
Châtillon Cedex, 92323
France
Tel: +33 1 49 65 2500

Asia
7 Changi South St. 2
#04-00
Singapore, 486415
Tel: +65 580 6000

www.ads.avnet.com